

ASYMPTOTIC EQUIVALENCE OF MINIMUM DISCREPANCY FUNCTION ESTIMATORS TO G.L.S. ESTIMATORS

Alexander Shapiro: Department of Mathematics and Applied
Mathematics, University of South Africa,
P.O. Box 392, PRETORIA, 0001, South Africa.

Key words: Asymptotic equivalence; covariance structures;
discrepancy function; generalized least
squares.

Summary: This paper is concerned with the asymptotic
theory of covariance or more generally moment
structural models. It is shown that for any
discrepancy function satisfying a few simple
axioms the corresponding estimators are asymp-
totically equivalent to GLS estimators.

1. INTRODUCTION

In the analysis of covariance structures, parameter estimates are usually obtained by minimizing a function measuring the discrepancy between the sample covariance matrix and a covariance matrix reproduced according to the model. Asymptotic distribution theory is readily available for the special case where the discrepancy function is a generalized least squares (GLS) function (e.g. Browne 1974, 1984; Lee & Bentler 1980; Shapiro 1983). This theory covers the widely used maximum likelihood (ML)

STMA 1985 SUBJECT CLASSIFICATION: 04:030

estimators which may also be considered as GLS estimators for an appropriately chosen weight matrix (Browne 1974, proposition 6). Other discrepancy functions, which do not fit into the GLS framework, can, however, also have desirable properties (e.g. Swain 1975). The present paper will show why there is no need to develop special asymptotic distribution theory for arbitrary discrepancy functions. It will be seen that for every discrepancy function satisfying mild conditions there corresponds a GLS discrepancy function which yields estimators and a test statistic with the same asymptotic distributions. Asymptotic distribution theory for any arbitrary minimum discrepancy function (MDF) estimators is then immediately available from the known asymptotic distribution theory for the corresponding GLS estimators.

2. STRUCTURAL MODELS AND MDF ESTIMATORS

In a moment structure, moments (usually of first and/or second order) are regarded as functions of a $q \times 1$ parameter vector θ , which belong to a specified parameter space Ω . That is the elements of the $p \times 1$ mean vector μ and/or $p(p+1)/2$ nonduplicated elements of the $p \times p$ covariance matrix Σ , forming the $m \times 1$ vector $\xi = (\xi_1, \dots, \xi_m)'$, are considered as functions of θ :

$$(2.1) \quad \xi_i = g_i(\theta), \quad i=1, \dots, m.$$

It can be said that equations (2.1) describe the particular structure (model) under investigation which relates the parameter vector θ to μ and Σ . The model is said to hold if there exists a vector $\theta_0 \in \Omega$ such that

$$(2.2) \quad \xi_0 = g(\theta_0),$$

where ξ_0 is the population value of ξ and $g(\theta) = (g_1(\theta), \dots, g_m(\theta))'$. Vector θ_0 is regarded as the true although unknown value (population value) of θ .

Given a sample estimate $\hat{x}$ of ξ_0 one fits the model (2.1) by minimizing the discrepancy between $\hat{x}$ and $\xi = g(\theta)$, which is measured by means of a certain real-valued function $F(x, \xi)$ of two $m \times 1$ vector variables x and ξ . Following the terminology proposed by Browne (1982) we call $F(x, \xi)$ the *discrepancy function*. The MDF estimator $\hat{\theta}$ of θ_0 is chosen to be a minimizer of $F(\hat{x}, g(\cdot))$ over the set Ω , i.e.

$$F(\hat{x}, g(\hat{\theta})) = \min_{\theta \in \Omega} F(\hat{x}, g(\theta)).$$

The MDF test statistic is given by $n\hat{F}$, where n is the sample size and $\hat{F}$ is the minimum of the discrepancy function $\hat{F} = F(\hat{x}, g(\hat{\theta}))$.

We suppose that the discrepancy function satisfies the following conditions (Browne 1982, p. 81):

- (i) $F(x, \xi) \geq 0$ for all x and ξ .
- (ii) $F(x, \xi) = 0$ if and only if $x = \xi$.
- (iii) F is twice continuously differentiable in x and ξ .

Conditions (i)-(iii) imply the following key result about the structure of discrepancy functions.

Theorem 1:

If $F(x, \xi)$ satisfies conditions (i)-(iii), then there exists a continuous $m \times m$ symmetric matrix valued function $V_{x, \xi} = V(x, \xi)$ of x and ξ such that

$$(2.3) \quad F(x, \xi) = (x - \xi)' V_{x, \xi} (x - \xi).$$

Proof:

Let us fix a point ξ and consider the function $f(x) = F(x, \xi)$. We also define the function $\varphi(t) = f(\xi + t(x - \xi))$ of parameter t . Clearly

$\varphi(0) = f(\xi)$ and $\varphi(1) = f(x)$. Then it follows from the fundamental (Newton-Leibnitz) theorem of calculus that

$$f(x) - f(\xi) = \varphi(1) - \varphi(0) = \int_0^1 \varphi'(t) dt,$$

with

$$\varphi'(t) = \sum_{i=1}^m (x_i - \xi_i) f'_i(\xi + t(x - \xi)).$$

Here $f'_i(z)$ denotes the first-order partial derivative of f with respect to the i -th argument at the point z . From (ii) we have that $f(\xi)$ is zero and hence

$$(2.4) \quad f(x) = \sum_{i=1}^m (x_i - \xi_i) \int_0^1 f'_i(\xi + t(x - \xi)) dt.$$

Now for fixed x, ξ and t consider the functions

$$\psi_i(u) = f'_i(\xi + ut(x - \xi)), \quad i=1, \dots, m.$$

Clearly $\psi_i(0) = f'_i(\xi)$ and $\psi_i(1) = f'_i(\xi + t(x - \xi))$. It follows from (i) and (ii) that $f(x)$ attains its minimal value (of zero) at the point ξ and hence $f'_i(\xi) = 0$, $i=1, \dots, m$. Applying again the fundamental theorem to the function $\psi_i(u)$ we obtain that

$$(2.5) \quad f'_i(\xi + t(x - \xi)) = \int_0^1 \psi'_i(u) du,$$

where

$$\psi'_i(u) = \sum_{j=1}^m t(x_j - \xi_j) f''_{ij}(\xi + ut(x - \xi))$$

and f''_{ij} denotes the second-order partial derivative. Substituting (2.5) into (2.4) we obtain

$$f(x) = \sum_{i,j=1}^m (x_i - \xi_i)(x_j - \xi_j) \int_0^1 \int_0^1 t f''_{ij}(\xi + ut(x - \xi)) du dt.$$

This implies (2.3) with the typical element of the (symmetric) matrix $V(x, \xi)$ given by

$$(2.6) \quad [V(x, \xi)]_{ij} = \int_0^1 \int_0^1 t f''_{ij}(\xi + ut(x - \xi)) du dt.$$

The matrix $V(x, \xi)$ defined by (2.6) depends continuously on x and ξ (see, e.g., Fikhtengolts 1965, p. 140, theorem 2).

The result of theorem 1 demonstrates that any discrepancy function, satisfying conditions (i)-(iii), can be represented in the form of an "extended" GLS function, where the word "extended" indicates that the corresponding weight matrix $V(x, g(\theta))$ may depend on θ as well as on x . Expression (2.6) gives a constructive way of reproducing this weight matrix although, of course, it is often impossible to calculate the required double integrals explicitly. One may note that if $F(x, \xi)$ admits k -th-order continuous partial derivatives, then the elements of $V(x, \xi)$ given by (2.6) are at least $k-2$ times differentiable functions of x and ξ .

Another important consequence of theorem 1 is that it explains the second-order structure of F . Namely, the second-order Taylor approximation of F at the (population) point (ξ_0, ξ_0) is $(x - \xi)' V_0 (x - \xi)$, where V_0 is the (matrix) value of $V(x, \xi)$ at (ξ_0, ξ_0) . Alternatively the matrix V_0 is given by any one of the Hessian matrices $\frac{1}{2} \partial^2 F / \partial x \partial x'$, $-\frac{1}{2} \partial^2 F / \partial x \partial \xi'$ or $\frac{1}{2} \partial^2 F / \partial \xi \partial \xi'$ calculated at the point (ξ_0, ξ_0) . This implies that the second-order Taylor approximation of F at (ξ_0, ξ_0) is *symmetric* in x and ξ while the discrepancy function itself is possibly not. Finally we note that the Hessian matrices of $F(x, g(\theta))$ at (ξ_0, θ_0) are now readily available. By the chain rule of differentiation we have that $\partial^2 F / \partial x \partial \theta'$ and $\partial^2 F / \partial \theta \partial \theta'$ are equal to $-2V_0 \Delta$ and $2\Delta' V_0 \Delta$ respectively, where $\Delta = \partial g / \partial \theta'$ denotes the $m \times q$ Jacobian matrix of g at θ_0 .

3. ASYMPTOTIC THEORY

Let $\hat{x}$ be a sample estimate of ξ_0 . In this section we investigate the (asymptotic) distribution of the corresponding MDF estimator $\hat{\theta}$ and the test statistic $n\hat{F}$. We suppose throughout that $n^{\frac{1}{2}}(\hat{x}-\xi_0)$ has asymptotically a multivariate normal distribution with a null mean vector and a certain covariance matrix Γ . Also the following regularity conditions will be required:

- (R1) The parameter space Ω is compact.
- (R2) The parameter vector θ is identified at θ_0 , i.e. $g(\theta^*) = g(\theta_0)$ and $\theta^* \in \Omega$ implies that $\theta^* = \theta_0$.
- (R3) θ_0 is an interior point of Ω .
- (R4) The $q \times q$ matrix $\Delta'V_0\Delta$ is nonsingular.

A few remarks may clarify our regularity conditions and their implications. Conditions (R1) and (R2) alone imply the consistency of the MDF estimator $\hat{\theta}$ (see Shapiro 1984). Condition (R4) implies that the matrix Δ is non-degenerate, i.e. is of full rank q . Moreover, if V_0 is nonsingular (and hence is positive definite), then (R4) is equivalent to the non-degeneracy of Δ .

Now let us consider the following GLS discrepancy function

$$G(x, \xi) = (x - \xi)'W(x - \xi),$$

where W is a symmetric matrix valued function of x converging to the matrix V_0 as x tends to ξ_0 . We denote by $\tilde{\theta}$ the corresponding GLS estimator of θ_0 and by $\tilde{G}$ the minimal value $\tilde{G} = G(\hat{x}, g(\tilde{\theta}))$. The discrepancy function $G(x, \xi)$ is close to $F(x, \xi)$, given in the form (2.3), in the sense that it has the same second-order Taylor approximation at the point (ξ_0, ξ_0) . It follows from the standard asymptotic theory that both $n^{\frac{1}{2}}(\hat{\theta} - \theta_0)$ and $n^{\frac{1}{2}}(\tilde{\theta} - \theta_0)$ are asymptotically equal to

$$n^{\frac{1}{2}}(\Delta'V_0\Delta)^{-1}\Delta'V_0(\hat{x}-\xi_0).$$

Also $n\hat{F}$ and $n\tilde{G}$ are asymptotically equal to

$$n(\hat{x}-\xi_0)'U(\hat{x}-\xi_0)$$

with

$$U = V_0 - V_0\Delta(\Delta'V_0\Delta)^{-1}\Delta'V_0$$

(see Browne 1974, Shapiro 1983).

This implies the following result:

Theorem 2:

Suppose that regularity conditions (R1)-(R4) hold. Then

$$n^{\frac{1}{2}}(\hat{\theta}-\theta_0) \stackrel{a}{=} n^{\frac{1}{2}}(\tilde{\theta}-\theta_0)$$

and

$$n\hat{F} \stackrel{a}{=} n\tilde{G}.$$

The notation " $\stackrel{a}{=}$ " is read "is asymptotically equal to" and means that the difference between the left and right sides approaches zero in probability as the sample size tends to infinity.

Consider for example the ML discrepancy function

$$F_{ML}(S, \Sigma) = \log |\Sigma| - \log |S| + \text{tr}(S\Sigma^{-1}) - p.$$

The function F_{ML} can be considered as a function of two vectors $s = \text{vec}(S)$ and $\sigma = \text{vec}(\Sigma)$, where $\text{vec}(S)$ denotes the $p^2 \times 1$ vector formed from the elements of S taken columnwise. It is well known that the Hessian matrix $\partial^2 F_{ML} / \partial s \partial s'$ is equal to $S^{-1} \otimes S^{-1}$, where $\otimes$ denotes the Kronecker product of matrices. Then we obtain from theorem 1 that

$$F_{ML}(S, \Sigma) = (s - \sigma)' V (s - \sigma)$$

with the matrix V given by

$$V = \int_0^1 \int_0^1 t(\Sigma + ut(S - \Sigma))^{-1} \otimes (\Sigma + ut(S - \Sigma))^{-1} du dt.$$

As S tends to Σ , the matrix V tends to

$$\frac{1}{2} \frac{\partial^2 F_{ML}}{\partial s \partial s'} \Big|_{S=\Sigma} = \frac{1}{2} \Sigma^{-1} \otimes \Sigma^{-1}.$$

This implies the well-known result (Browne 1974) that the ML approach is asymptotically equivalent to the GLS method with the discrepancy function $\frac{1}{2} \text{tr} [(S - \Sigma) S^{-1}]^2$. Also, as we have mentioned, the second-order structure of a discrepancy function is symmetric in S and Σ . Therefore the same asymptotics will be given by the discrepancy functions $\frac{1}{2} \text{tr} [(\Sigma - S) \Sigma^{-1}]^2$ and $\log |S| - \log |\Sigma| + \text{tr}(\Sigma S^{-1}) - p$ as well.

ACKNOWLEDGEMENT

The author is indebted to Professor M.W. Browne for many constructive comments and helpful suggestions.

REFERENCES

- BROWNE, M.W. (1974). Generalized least squares estimators in the analysis of covariance structures. *South African Statist. J.* **8**, 1-24.
- BROWNE, M.W. (1982). Covariance structures. In: D.M. Hawkins (ed.). *Topics in applied multivariate analysis*. Cambridge University Press.
- BROWNE, M.W. (1984). Asymptotically distribution-free methods for the analysis of covariance structures. *Brit. J. Math. Statist. Psychol.* **37**, 62-83.

- FIKHTENGOLTS, G.M. (1965). *The fundamentals of mathematical analysis*. Vol. II, New York: Pergamon Press.
- LEE, S.Y. & BENTLER, P.M. (1980). Some asymptotic properties of constrained generalized least squares estimation in covariance structure models. *South African Statist. J.* **14**, 121-136.
- SHAPIRO, A. (1983). Asymptotic distribution theory in the analysis of covariance structures (a unified approach). *South African Statist. J.* **17**, 33-81.
- SHAPIRO, A. (1984). A note on the consistency of estimators in the analysis of moment structures. *Brit. J. Math. Statist. Psychol.* **37**, 84-88.
- SWAIN, A.J. (1975). A class of factor analysis estimation procedures with common asymptotic sampling properties. *Psychometrika* **40**, 315-335.